

Part I

SUMMARY of Applications of Integrals

Summary

February 21, 2026

SUMMARY: Velocity, Speed, Position, Displacement, Distance

Position, $s(t)$, of an object at time t : $s(t) = s(a) + \int_a^t v(z)dz$

Displacement, $\Delta s = s(b) - s(a)$,
of an object over the time interval $[a, b]$: $s(b) - s(a) = \int_a^b v(t)dt$

Distance traveled by an object
over the time interval $[a, b]$: $\int_a^b |v(t)|dt$

SUMMARY: Rate of Accumulation, Amount, Change in the Amount

The amount, $A(t)$, of some
substance/population at the time t : $A(t) = A(a) + \int_a^t A'(z)dz$

The amount, $A(b)$,
over the time interval $[a, b]$: $A(b) = A(a) + \int_a^b A'(t)dt$

The change in the amount, $A(b) - A(a)$,
over the time interval $[a, b]$: $A(b) - A(a) = \int_a^b A'(t)dt$

SUMMARY: Average Value, the Mean Value Theorem for Integrals

Average value $\bar{f}$,
of the function f on the interval $[a, b]$: $\bar{f} = \frac{1}{b-a} \int_a^b f(x)dx$

Mean Value Theorem for Integrals

Let f be continuous on $[a, b]$. There exists a value c in (a, b) such that

Summary

$$\frac{1}{b-a} \int_a^b f(x) dx = f(c)$$

Note 1: $f(c) = \bar{f}$, the average value of f on $[a, b]$.

Note 2: The net area of the region between the curve $y = f(x)$ and the x -axis is given by

$$\int_a^b f(x) dx = f(c)(b-a)$$

Part II

Recitation Questions

Problem 1

February 21, 2026

Problem 1 Solve the following word problems:

(a) The velocity function for an object moving along a line east/west is given by $v(t) = -t^2 + 4t - 3$ feet per minute.

(i) Find the total displacement the object traveled from 2 minutes to 6 minutes (assume east is positive).

Explanation. The object's total displacement is given by $\int_2^6 v(t) dt$. So we compute:

$$\begin{aligned}\int_2^6 v(t) dt &= \int_2^6 (-t^2 + 4t - 3) dt \\ &= \left[-\frac{1}{3}t^3 + 2t^2 - 3t \right]_2^6 \\ &= (-72 + 72 - 18) - \left(-\frac{8}{3} + 8 - 6 \right) \\ &= \frac{8}{3} - 20 = -\frac{52}{3}.\end{aligned}$$

So the object's displacement is $\frac{52}{3}$ feet west of its original location.

(ii) Find the total distance the object traveled from 2 minutes to 6 minutes.

Explanation. First notice that $v(t) = -(t^2 - 4t + 3) = -(t-1)(t-3)$. So we can see that

$$v(t) > 0 \text{ when } 2 \leq t < 3.$$

$$v(t) < 0 \text{ when } 3 < t \leq 6.$$

Thus, the total distance that the object traveled from 2 minutes to

6 minutes is:

$$\begin{aligned}
 \int_2^6 |v(t)| \, dt &= \int_2^3 |v(t)| \, dt + \int_3^6 |v(t)| \, dt \\
 &= \int_2^3 v(t) \, dt + \int_3^6 -v(t) \, dt \\
 &= \int_2^3 (-t^2 + 4t - 3) \, dt - \int_3^6 (-t^2 + 4t - 3) \, dt \\
 &= \left[-\frac{1}{3}t^3 + 2t^2 - 3t \right]_2^3 - \left[-\frac{1}{3}t^3 + 2t^2 - 3t \right]_3^6 \\
 &= \left((-9 + 18 - 9) - \left(-\frac{8}{3} + 8 - 6\right) \right) - \\
 &\quad ((-72 + 72 - 18) - (-9 + 18 - 9)) \\
 &= \left(0 - 2 + \frac{8}{3} \right) - (-18 - 0) \\
 &= 16 + \frac{8}{3} = \frac{56}{3}.
 \end{aligned}$$

So, the total distance that the object traveled is $\frac{56}{3}$ feet.

- (iii) Suppose that the object's position 2 minutes into the trip is 5 feet of a placement marker. What is its position (relative to the placement marker) at 6 minutes.

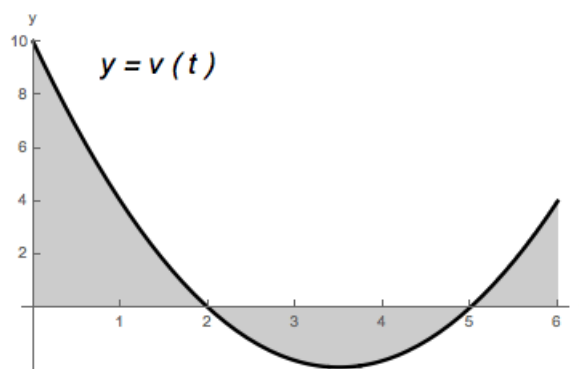
Explanation. $s(6) = s(2) + \int_2^6 v(t) \, dt = 5 + \left(-\frac{52}{3}\right) = -\frac{37}{3}.$

So the object's position at 6 minutes is $\frac{37}{3}$ feet west of the placement marker.

- (b) Sammy the Snail sets up camp in the median of I-70 and, starting at noon and ending at 6pm, hikes back and forth along the highway. He starts his hike at his campsite. His velocity at time t hours (after noon) is given by $v(t) = (t - 2)(t - 5)$ inches per hour. Find the total distance Sammy travelled on his hike.

Explanation. The total distance that Sammy travels is $\int_0^6 |v(t)| \, dt$. The following picture indicates where $v(t)$ is positive and negative:

Problem 1



So we compute:

$$\begin{aligned}
 \int_0^6 |v(t)| \, dt &= \int_0^2 |v(t)| \, dt + \int_2^5 |v(t)| \, dt + \int_5^6 |v(t)| \, dt \\
 &= \int_0^2 v(t) \, dt - \int_2^5 v(t) \, dt + \int_5^6 v(t) \, dt \\
 &= \int_0^2 (t^2 - 7t + 10) \, dt - \int_2^5 (t^2 - 7t + 10) \, dt + \int_5^6 (t^2 - 7t + 10) \, dt \\
 &= \left[\frac{1}{3}t^3 - \frac{7}{2}t^2 + 10t \right]_0^2 - \left[\frac{1}{3}t^3 - \frac{7}{2}t^2 + 10t \right]_2^5 + \left[\frac{1}{3}t^3 - \frac{7}{2}t^2 + 10t \right]_5^6 \\
 &= \left(\left(\frac{8}{3} - 14 + 20 \right) - 0 \right) - \left(\left(\frac{125}{3} - \frac{175}{2} + 50 \right) - \left(\frac{8}{3} + 6 \right) \right) + \\
 &\quad \left((72 - 126 + 60) - \left(\frac{125}{3} - \frac{175}{2} + 50 \right) \right) \\
 &= -82 + 175 - 78 = 15.
 \end{aligned}$$

So Sammy has traveled a distance of 15 inches.

Problem 2

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Problem 2 Suppose that $r(t) = r_0 e^{-kt}$ (with $k > 0$) is the rate at which a nation extracts oil. The current rate of extraction is $r(0) = 10^7$ barrels/yr. Also assume that the estimate of the total oil reserve (ie, the amount of oil remaining beneath the ground in this country) is 2×10^9 barrels.

- (a) Find $A(t)$, the total amount of oil extracted by the nation after t years.

Explanation. $A(t) = \int_0^t r(s) ds$

$$\begin{aligned} A(t) &= \int_0^t r(s) ds \\ &= \int_0^t r_0 e^{-ks} ds \\ &= -\frac{r_0}{k} \left[e^{-ks} \right]_0^t \\ &= -\frac{r_0}{k} (e^{-kt} - 1) \\ &= -\frac{1}{k} 10^7 (e^{-kt} - 1) \end{aligned}$$

- (b) Evaluate $\lim_{t \rightarrow \infty} A(t)$ and explain the meaning of this limit.

Explanation.

$$\begin{aligned} \lim_{t \rightarrow \infty} A(t) &= \lim_{t \rightarrow \infty} -\frac{1}{k} 10^7 (e^{-kt} - 1) \\ &= -\frac{1}{k} 10^7 (0 - 1) \\ &= \frac{1}{k} 10^7. \end{aligned}$$

- (c) Find the minimum constant k for which the total oil reserves will last forever.

Explanation. For the oil reserves to last forever, we need that

$$\begin{aligned} \lim_{t \rightarrow \infty} A(t) &\leq 2 \times 10^9 \\ \iff \frac{1}{k} 10^7 &\leq 2 \times 10^9 \\ \iff \frac{1}{k} &\leq 2 \times 10^2 = 200 \\ \iff \frac{1}{200} &\leq k. \end{aligned}$$

So the minimum value for k is $\frac{1}{200}$.

Problem 2

- (d) Suppose that the constant k is half the minimum value found in part (c). When will the nation deplete its oil reserve?

Explanation. First note that $k = \frac{1}{2} \cdot \frac{1}{200} = \frac{1}{400}$. We want to find the value of t such that:

$$A(t) = 2 \times 10^9$$

$$-400 \times 10^7 \left(e^{-\frac{1}{400}t} - 1 \right) = 2 \times 10^9$$

$$\left(e^{-\frac{1}{400}t} - 1 \right) = \frac{2 \times 10^9}{-400 \times 10^7} = -\frac{1}{2}$$

$$e^{-\frac{1}{400}t} = \frac{1}{2}$$

$$-\frac{1}{400}t = \ln\left(\frac{1}{2}\right) = -\ln(2)$$

$$t = 400 \ln(2) \approx 277.259 \text{ years.}$$

Problem 3

February 21, 2026

Problem 3 Assume that the rate of change (in dollars per day) of the price of shares of stock in the WeSaySo Company (with t in days) is modeled by the equation $r(t) = -3t^2 + 30t - 63$ (note that this is technically a discrete function, but prices change so often with stocks that modeling this with a continuous function makes sense). Assume also that the price of a share of stock on day 1 (i.e., $t = 1$) is \$51. Answer the following questions:

- (a) Find the rate of change of price at $t = 5$.

Explanation. $r(5) = -3(25) + 30(5) - 63 = -75 + 150 - 63 = 12$ dollars/day.

- (b) Find the price of a share of stock at $t = 5$.

Explanation. Let $p(t)$ denote the price of a share of stock at any time t . Then notice that $r(t) = p'(t)$. We will first find $p(t)$ for general t , and then substitute $t = 5$ to solve this problem.

$$\begin{aligned} p(t) &= \int_1^t r(s) ds + p(1) \\ &= \int_1^t (-3s^2 + 30s - 63) ds + 51 \\ &= \left[-s^3 + 15s^2 - 63s \right]_1^t + 51 \\ &= (-t^3 + 15t^2 - 63t) - (-1 + 15 - 63) + 51 \\ &= -t^3 + 15t^2 - 63t + 100. \end{aligned}$$

So, $p(5) = -125 + 375 - 315 + 100 = 35\$$.

- (c) How fast is the rate of change of price changing at $t = 5$?

Explanation. $r'(t) = -6t + 30$. So, $r'(5) = -30 + 30 = 0$.

- (d) How much did the price of a share of stock change in the first 6 days (i.e., on $[1, 6]$)?

Explanation. $p(6) - p(1) = (-216 + 540 - 378 + 100) - 51 = -5\$$.

- (e) What was the greatest rate of change of price during the first 6 days (i.e., on $[1, 6]$)? [3]

Explanation. This question wants us to maximize $r(t)$ on the closed interval $[1, 6]$. So we need to find all critical points of $r(t)$ on $[1, 6]$.

$$r'(t) = -6t + 30 := 0 \implies t = 5$$

Then, using the closed interval method, we simply plug $t = 1, 5, 6$ into $r(t)$ and check which has the greatest output.

$$r(1) = -3 + 30 - 63 = -36$$

$$r(5) = -75 + 150 - 63 = 12$$

$$r(6) = -108 + 180 - 63 = 9.$$

Problem 3

Thus, the greatest rate of change of price is 12 dollars/day when $t = 5$.

- (f) What was the greatest price of a share of stock during the first 6 days (i.e., on $[1, 6]$)? [3]

Explanation. This question wants us to maximize $p(t)$ on the closed interval $[1, 6]$. So we need to find all critical points of $p(t)$ on $[1, 6]$. But note that $p'(t) = r(t)$. So critical points of $p(t)$ are just roots of $r(t)$.

$$r(t) = 0$$

$$-3t^2 + 30t - 63 = 0$$

$$-3(t^2 - 10t + 21) = 0$$

$$-3(t - 3)(t - 7) = 0$$

$$t = 3, 7 \implies t = 3$$

since 7 is not in the interval $[1, 6]$. So we use the Closed Interval Method:

$$p(1) = -1 + 15 - 63 + 100 = 51$$

$$p(3) = -27 + 135 - 189 + 100 = 19$$

$$p(6) = -216 + 540 - 378 + 100 = 46$$

Thus, the greatest price of the stock is \$51 when $t = 1$.

Problem 4

February 21, 2026

Problem 4 A cup of coffee has temperature $20 + 75e^{-.02t}$ degrees (Celsius) t minutes after being poured into a cup. What is the average temperature of the coffee during the first half hour?

Explanation. Let $T(t) := 20 + 75e^{-.02t}$. We want the average value of T on $[0, 30]$.

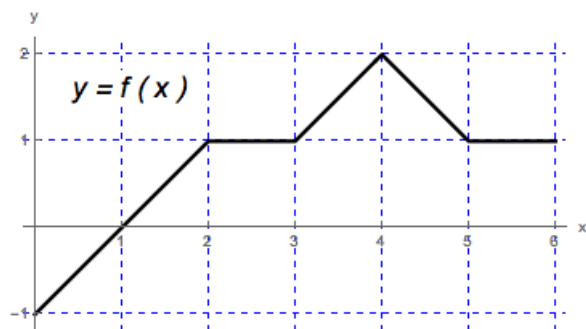
$$\begin{aligned} T_{\text{avg}} &= \frac{1}{30 - 0} \int_0^{30} (20 + 75e^{-.02t}) \, dt \\ &= \frac{1}{30} \left[20t - \frac{75}{.02} e^{-.02t} \right]_0^{30} \\ &= \frac{1}{30} \left[20t - 3750e^{-.02t} \right]_0^{30} \\ &= \frac{1}{30} [(20(30) - 3750e^{-0.6}) - (0 - 3750)] \\ &= \frac{1}{30} (4350 - 3750e^{-0.6}) \\ &= 145 - 125e^{-0.6} \end{aligned}$$

So the average temperature of the coffee during the first half hour is $145 - 125e^{-0.6} \approx 76.4$ degrees Celsius.

Problem 5

February 21, 2026

Problem 5 The graph of a function f defined on the interval $[0, 6]$ is given in the figure.



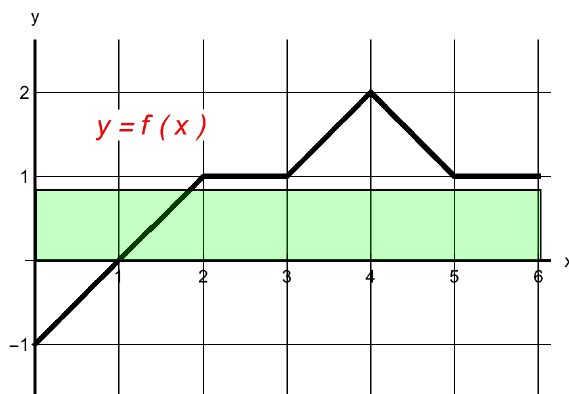
- (a) Compute the net area of the region between the graph of f and the x -axis, on the interval $[0, 6]$. [3]

Explanation. The net area $= -1/2 + 1/2 + 1 + 3 + 1 = 5$

- (b) Draw a rectangle with base on the x -axis, $0 \leq x \leq 6$, whose area is equal to the net area in part (a).

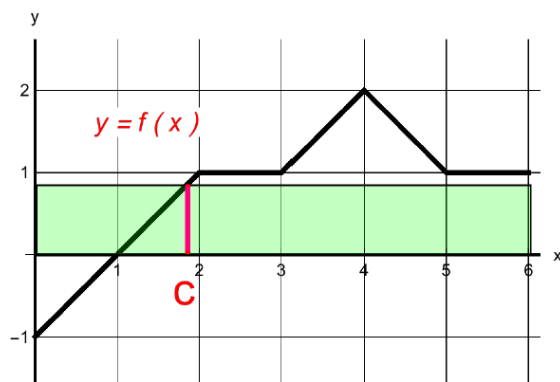
Explanation. Let A be the area of the rectangle. Then $A = (\text{base})(\text{height}) = 6h = 5$.

$$\text{So, } 6h = 5 \implies h = \frac{5}{6}$$



- (c) In the figure, mark a point c in $(0, 6)$ such that $f(c)$ is the height of the rectangle from part (b).

Explanation. $f(c) = h = \frac{5}{6}$



Note: In this example, there is only one such point c . In some cases there may be more than one.

- (d) Using the figure and parts (a-c), what is the relationship between the rectangle from part (b), the net area from part (a), and the average value of f on $[0, 6]$? [3]

Explanation. The Mean Value Theorem for integrals states that there exists a point d in (a, b) such that

$$f(d) = \frac{1}{b-a} \int_a^b f(x) dx.$$

The right-hand side of this equation is the average value of f on $[a, b]$.

Rewriting we have

$$f(d) \cdot (b-a) = \int_a^b f(x) dx.$$

The right-hand side of this equation is the net area found in part (a):

$$\int_0^6 f(x) dx = 5.$$

The left-hand side of this equation is the area of the rectangle in part (b).

The point d is the point c we found in part (c). From (b) and (c), $f(c) = \frac{5}{6}$ is the average value of f on $[0, 6]$.

Problem 6

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Problem 6 Find all points at which the given function equals its average value on the given interval.

(a) $f(x) = e^x$ $[0, 4]$

Explanation. First, we need to find f_{avg} :

$$\begin{aligned} f_{\text{avg}} &= \frac{1}{4-0} \int_0^4 e^x dx \\ &= \frac{1}{4} \left[e^x \right]_0^4 \\ &= \frac{1}{4} (e^4 - 1) \end{aligned}$$

So we are looking for all values $c \in [0, 4]$ such that:

$$\begin{aligned} f(c) &= \frac{1}{4} (e^4 - 1) \\ \implies e^c &= \frac{1}{4} (e^4 - 1) \\ \implies c &= \ln \left(\frac{1}{4} (e^4 - 1) \right) \end{aligned}$$

Therefore, our answer is $\ln \left(\frac{1}{4} (e^4 - 1) \right)$.

(b) $g(x) = \frac{\pi}{4} \sin(x)$ $[0, \pi]$

Explanation. First, we need to find g_{avg} :

$$\begin{aligned} g_{\text{avg}} &= \frac{1}{\pi - 0} \int_0^\pi \frac{\pi}{4} \sin(x) dx \\ &= \frac{1}{4} \left[-\cos(x) \right]_0^\pi \\ &= \frac{1}{4} (-(-1 - 1)) \\ &= \frac{1}{2} \end{aligned}$$

So we are looking for all values $c \in [0, \pi]$ such that:

$$\begin{aligned} g(c) &= \frac{1}{2} \\ \implies \frac{\pi}{4} \sin(c) &= \frac{1}{2} \\ \implies \sin(c) &= \frac{2}{\pi} \\ \implies c &= \arcsin \left(\frac{2}{\pi} \right), \pi - \arcsin \left(\frac{2}{\pi} \right) \end{aligned}$$

Problem 6

Therefore, our two answers are $\arcsin\left(\frac{2}{\pi}\right), \pi - \arcsin\left(\frac{2}{\pi}\right)$.

Problem 7

February 21, 2026

Problem 7 Find the average value of the function $g(t) = 4te^{-t^2}$ on the interval $[0, 3]$.

Explanation. We know $g_{\text{avg}} = \frac{1}{b-a} \int_a^b g(t) dt$ with $[a, b]$ as $[0, 3]$. We'll make a substitution, $w = t^2$ with $dw = 2t dt$. Notice that when $t = 0$, $w = (0)^2 = 0$ and when $t = 3$, $w = (3)^2 = 9$.

$$\begin{aligned} g_{\text{avg}} &= \frac{1}{3-0} \int_0^3 4te^{-t^2} dt \\ &= \frac{1}{3} \int_0^9 2e^{-w} dw \\ &= \frac{1}{3} \left[-2e^{-w} \right]_0^9 \\ &= \frac{1}{3} [(-2e^{-9}) - (-2e^0)] \\ &= \frac{1}{3} \left(2 - \frac{2}{e^9} \right) \end{aligned}$$